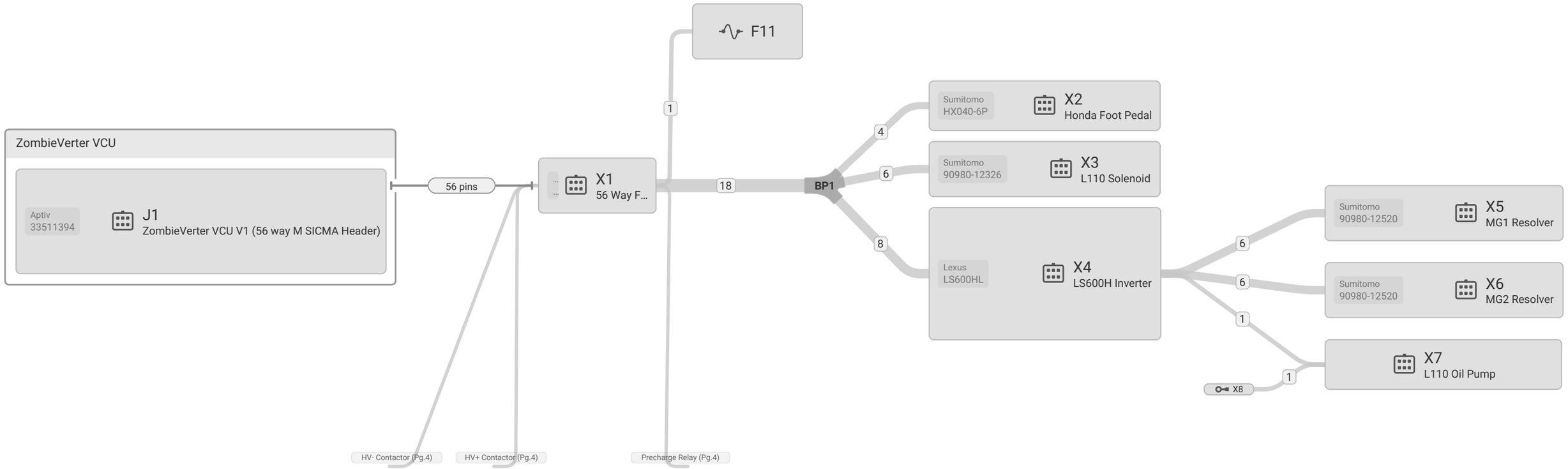
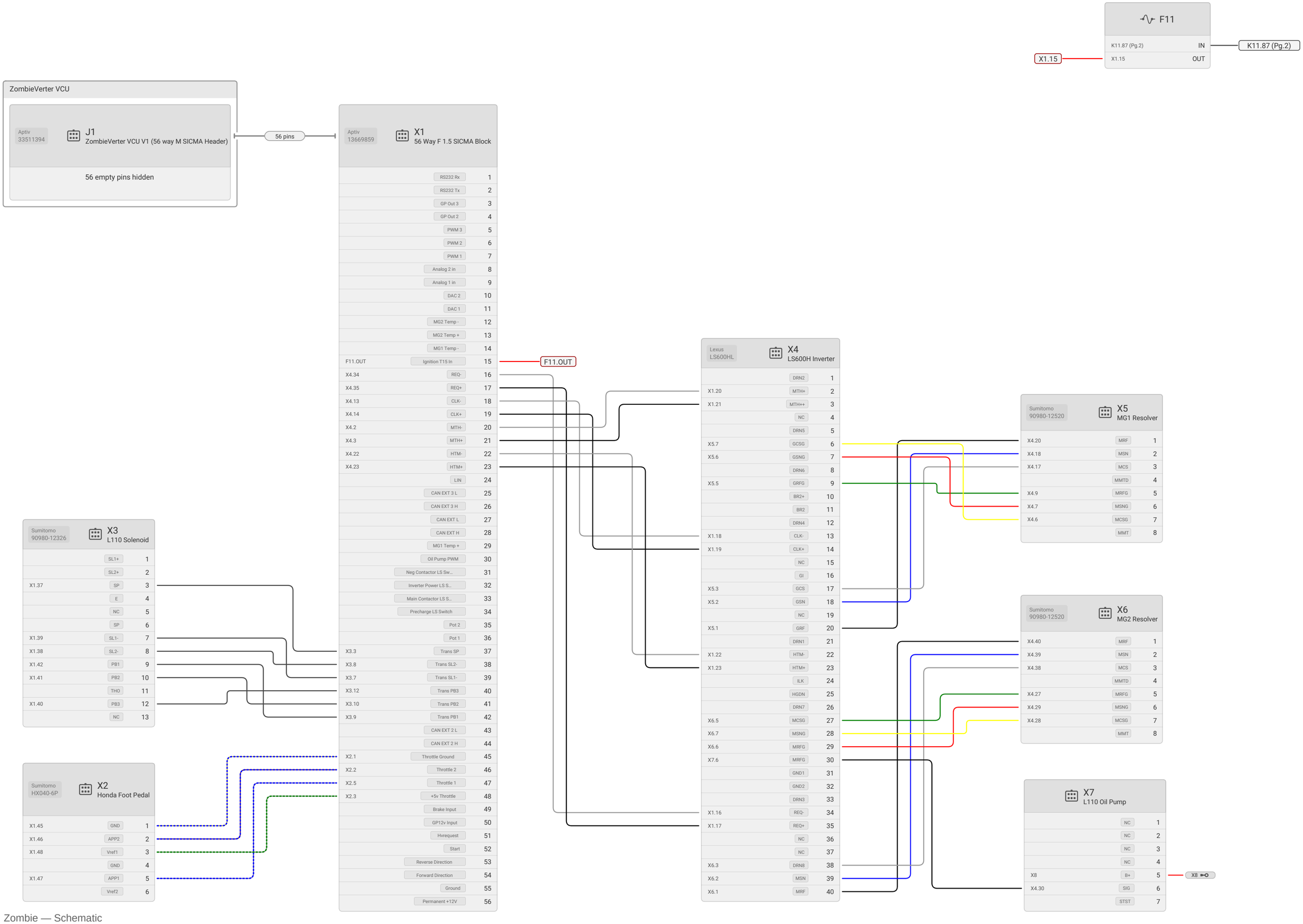
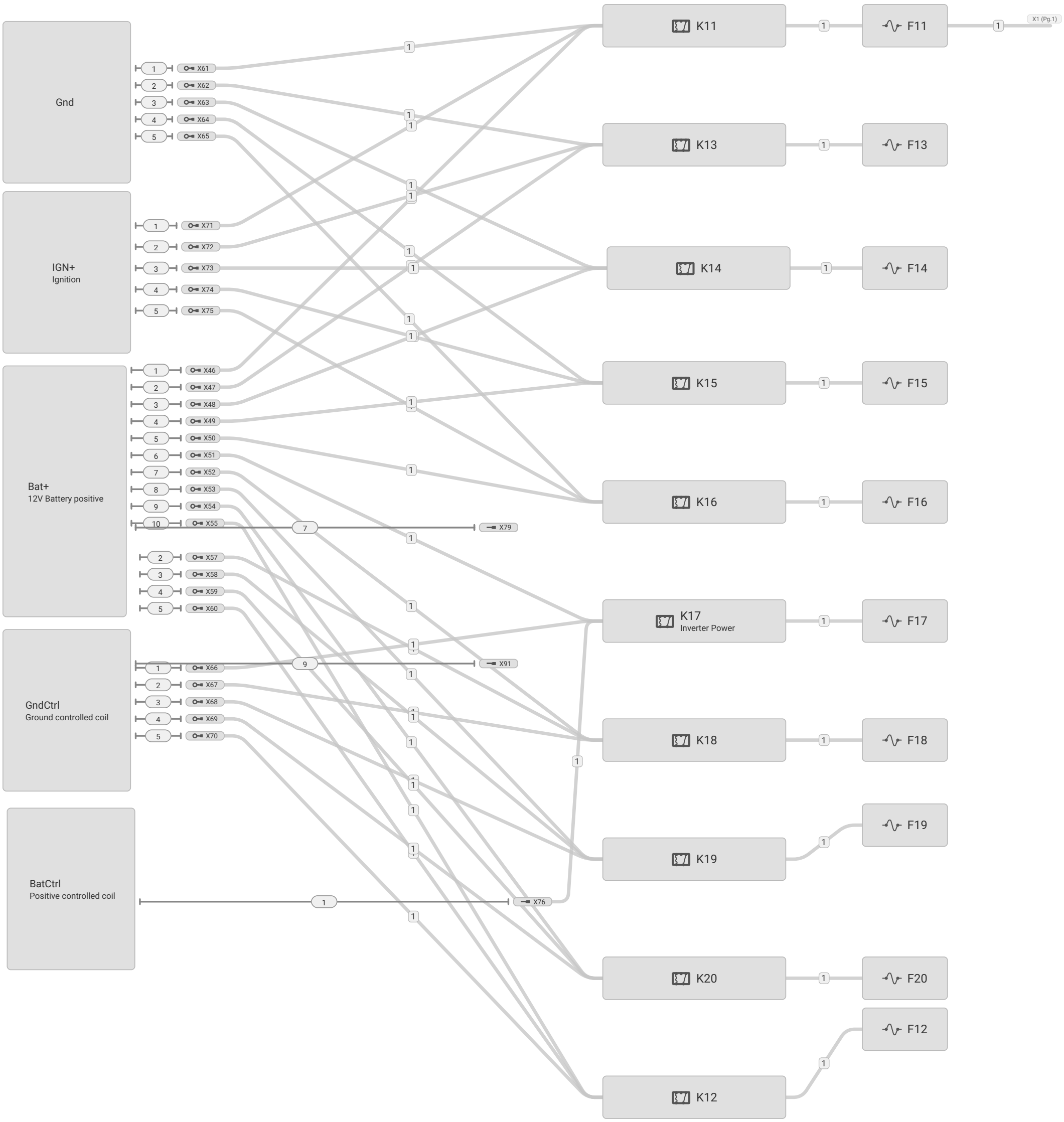
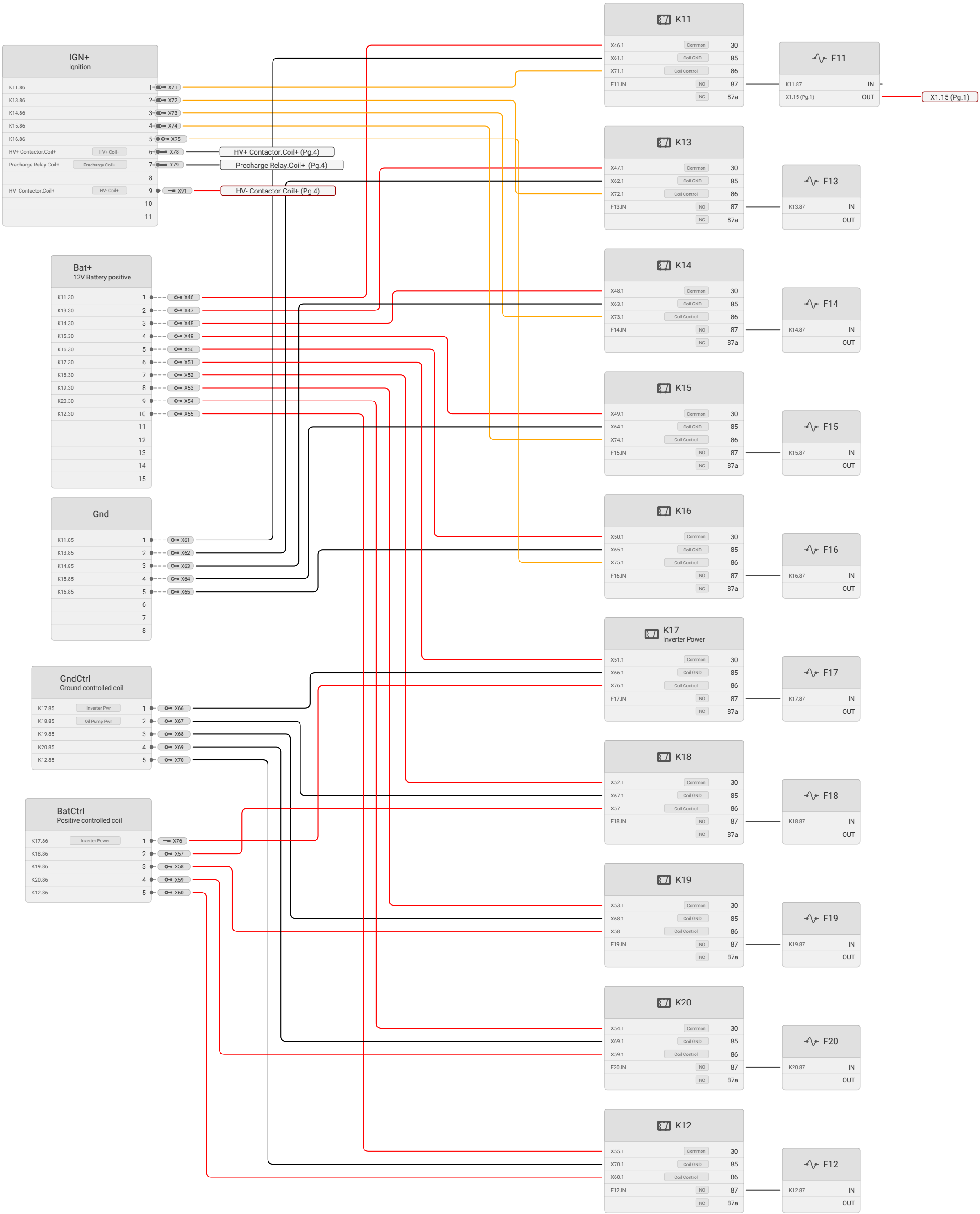
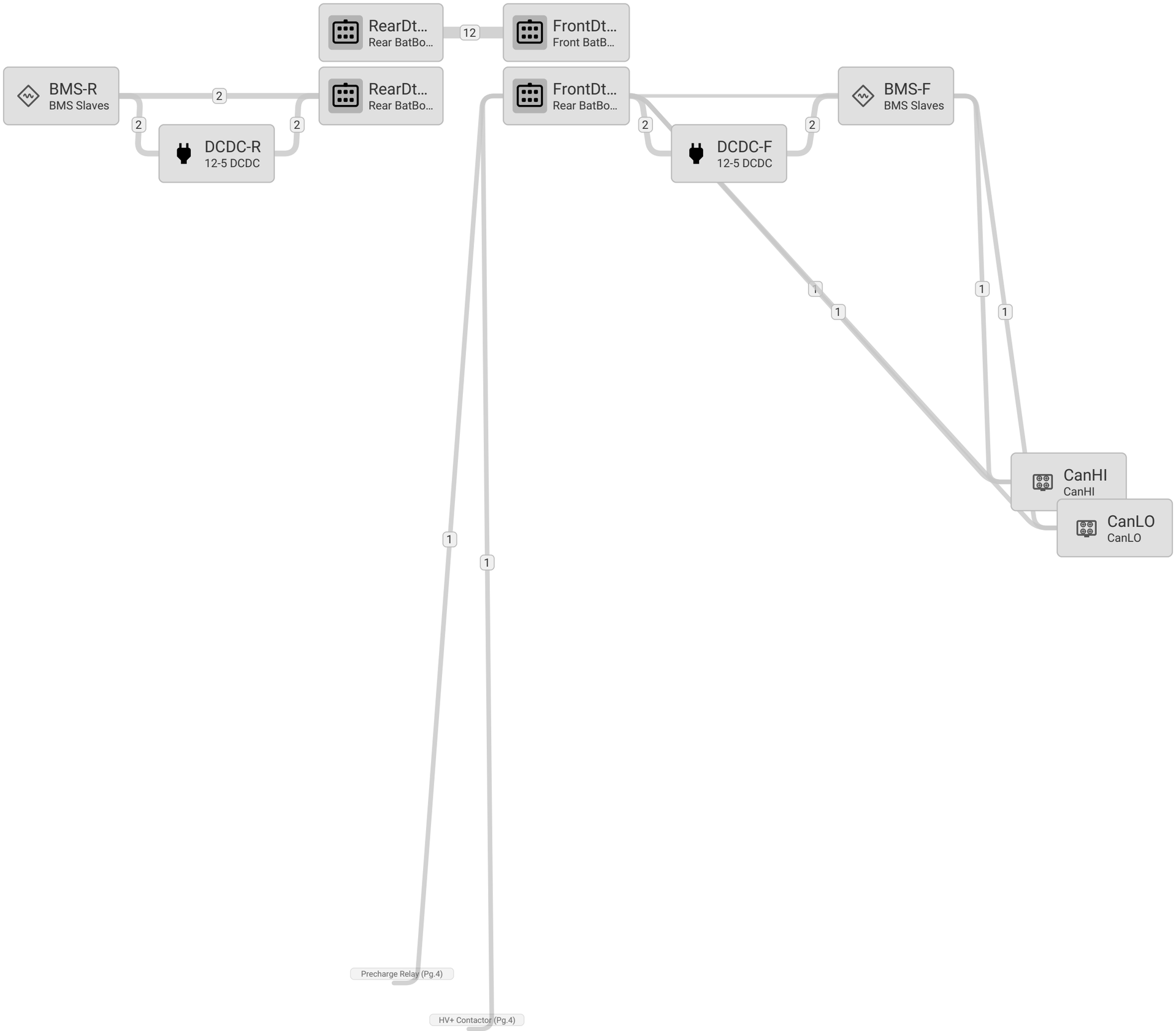

ZombieVerter VCU
Aptiv
33511394
J1
ZombieVerter VCU V1 (56 way M SICMA Header)
56 pins
X1
56 Way F...
F11
1
18
BP1
4
6
8
Sumitomo
HX040-6P
X2
Honda Foot Pedal
Sumitomo
90980-12326
X3
L110 Solenoid
Lexus
LS600HL
X4
LS600H Inverter
6
6
1
Sumitomo
90980-12520
X5
MG1 Resolver
Sumitomo
90980-12520
X6
MG2 Resolver
X7
L110 Oil Pump
1
X8
HV- Contactor (Pg.4)
HV+ Contactor (Pg.4)
Precharge Relay (Pg.4)


F11
K11.87 (Pg.2) IN
X1.15 OUT
X1.15
K11.87 (Pg.2)
ZombieVerter VCU
Aptiv 33511394
J1
ZombieVerter VCU V1 (56 way M SICMA Header)
56 empty pins hidden
56 pins
Aptiv 13669859
X1
56 Way F 1.5 SICMA Block
1 RS232 Rx
2 RS232 Tx
3 GP Out 3
4 GP Out 2
5 PWM 3
6 PWM 2
7 PWM 1
8 Analog 2 in
9 Analog 1 in
10 DAC 2
11 DAC 1
12 MG2 Temp -
13 MG2 Temp +
14 MG1 Temp -
F11.OUT 15 Ignition T15 In
F11.OUT
X4.34 16 REQ-
X4.35 17 REQ+
X4.13 18 CLK-
X4.14 19 CLK+
X4.2 20 MTH-
X4.3 21 MTH+
X4.22 22 HTM-
X4.23 23 HTM+
24 LIN
25 CAN EXT 3 L
26 CAN EXT 3 H
27 CAN EXT L
28 CAN EXT H
29 MG1 Temp +
30 Oil Pump PWM
31 Neg Contactor LS Sw...
32 Inverter Power LS S...
33 Main Contactor LS S...
34 Precharge LS Switch
35 Pot 2
36 Pot 1
X3.3 37 Trans SP
X3.8 38 Trans SL2-
X3.7 39 Trans SL1-
X3.12 40 Trans PB3
X3.10 41 Trans PB2
X3.9 42 Trans PB1
43 CAN EXT 2 L
44 CAN EXT 2 H
X2.1 45 Throttle Ground
X2.2 46 Throttle 2
X2.5 47 Throttle 1
X2.3 48 +5v Throttle
49 Brake Input
50 GP12v Input
51 Hvrequest
52 Start
53 Reverse Direction
54 Forward Direction
55 Ground
56 Permanent +12V
Sumitomo 90980-12326
X3
L110 Solenoid
1 SL1+
2 SL2+
X1.37 3 SP
4 E
5 NC
6 SP
X1.39 7 SL1-
X1.38 8 SL2-
X1.42 9 PB1
X1.41 10 PB2
11 THO
X1.40 12 PB3
13 NC
Sumitomo HX040-6P
X2
Honda Foot Pedal
X1.45 1 GND
X1.46 2 APP2
X1.48 3 Vref1
4 GND
X1.47 5 APP1
6 Vref2
Lexus LS600HL
X4
LS600H Inverter
1 DRN2
X1.20 2 MTH+
X1.21 3 MTH++
4 NC
5 DRN5
X5.7 6 GCSG
X5.6 7 GSNG
8 DRN6
X5.5 9 GRFG
10 BR2+
11 BR2
12 DRN4
X1.18 13 CLK-
X1.19 14 CLK+
15 NC
16 GI
X5.3 17 GCS
X5.2 18 GSN
19 NC
X5.1 20 GRF
21 DRN1
X1.22 22 HTM-
X1.23 23 HTM+
24 ILK
25 HGDN
26 DRN7
X6.5 27 MCSG
X6.7 28 MSNG
X6.6 29 MRFG
X7.6 30 MRFG
31 GND1
32 GND2
33 DRN3
X1.16 34 REQ-
X1.17 35 REQ+
36 NC
37 NC
X6.3 38 DRN8
X6.2 39 MSN
X6.1 40 MRF
Sumitomo 90980-12520
X5
MG1 Resolver
X4.20 1 MRF
X4.18 2 MSN
X4.17 3 MCS
4 MMTD
X4.9 5 MRFG
X4.7 6 MSNG
X4.6 7 MCSG
8 MMT
Sumitomo 90980-12520
X6
MG2 Resolver
X4.40 1 MRF
X4.39 2 MSN
X4.38 3 MCS
4 MMTD
X4.27 5 MRFG
X4.29 6 MSNG
X4.28 7 MCSG
8 MMT
X7
L110 Oil Pump
1 NC
2 NC
3 NC
4 NC
X8 5 B+
X4.30 6 SIG
7 STST
X8

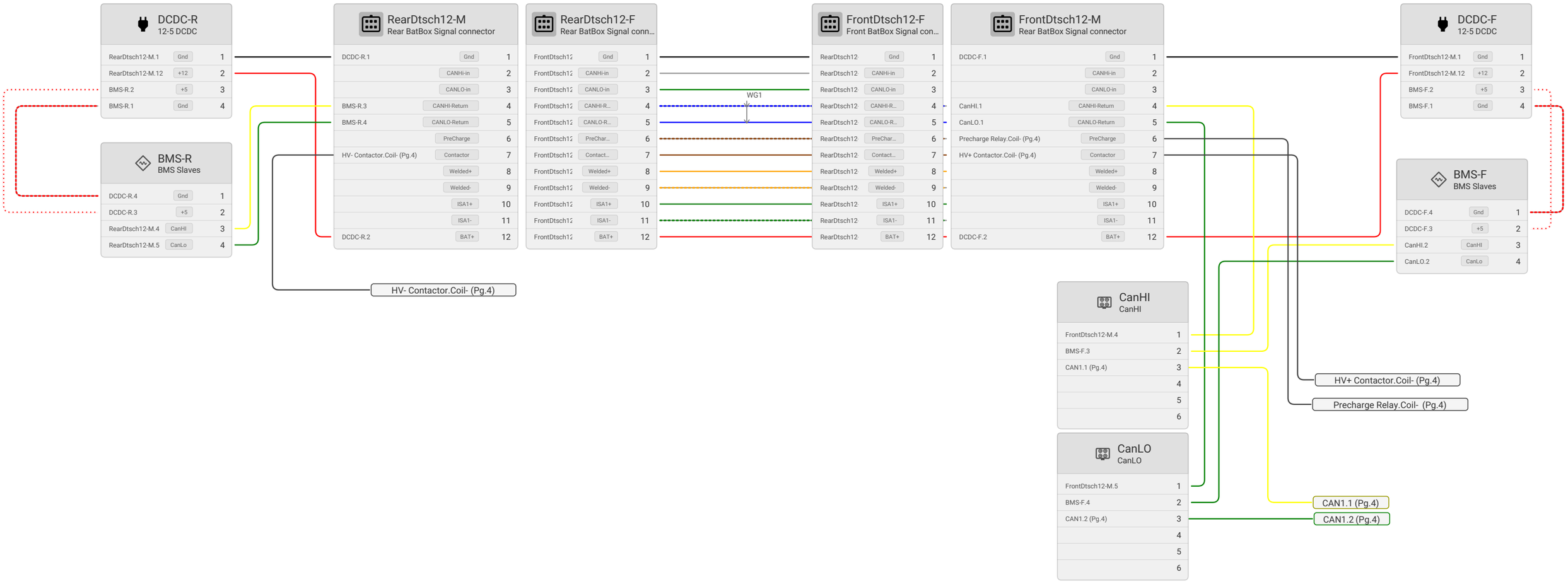

Gnd
IGN+
Ignition
Bat+
12V Battery positive
GndCtrl
Ground controlled coil
BatCtrl
Positive controlled coil
X61
X62
X63
X64
X65
X71
X72
X73
X74
X75
X46
X47
X48
X49
X50
X51
X52
X53
X54
X55
X57
X58
X59
X60
X66
X67
X68
X69
X70
X79
X91
X76
K11
K13
K14
K15
K16
K17
Inverter Power
K18
K19
K20
K12
F11
F13
F14
F15
F16
F17
F18
F19
F20
F12
X1 (Pg.1)

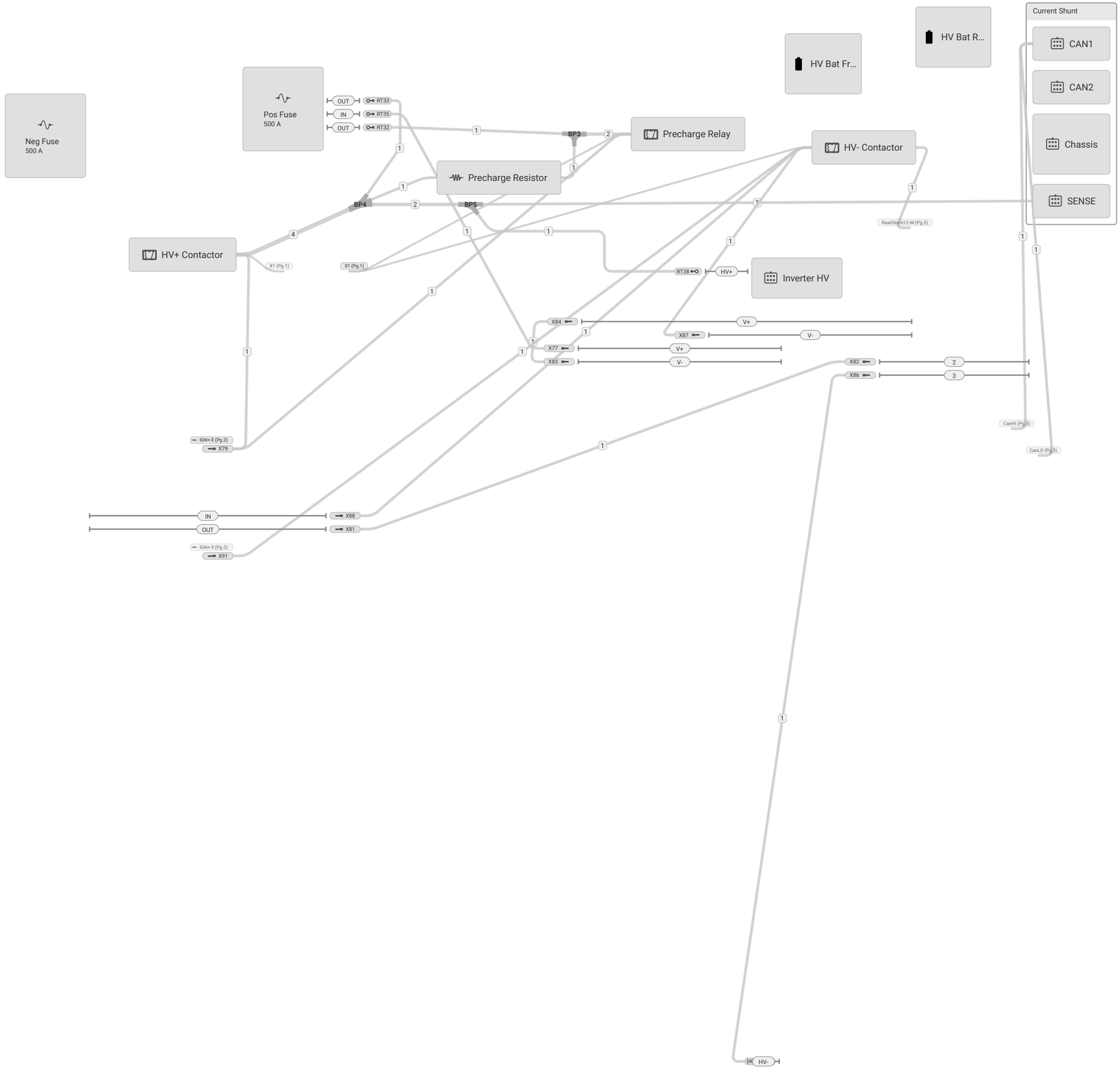


FuseRelay Panel — Schematic

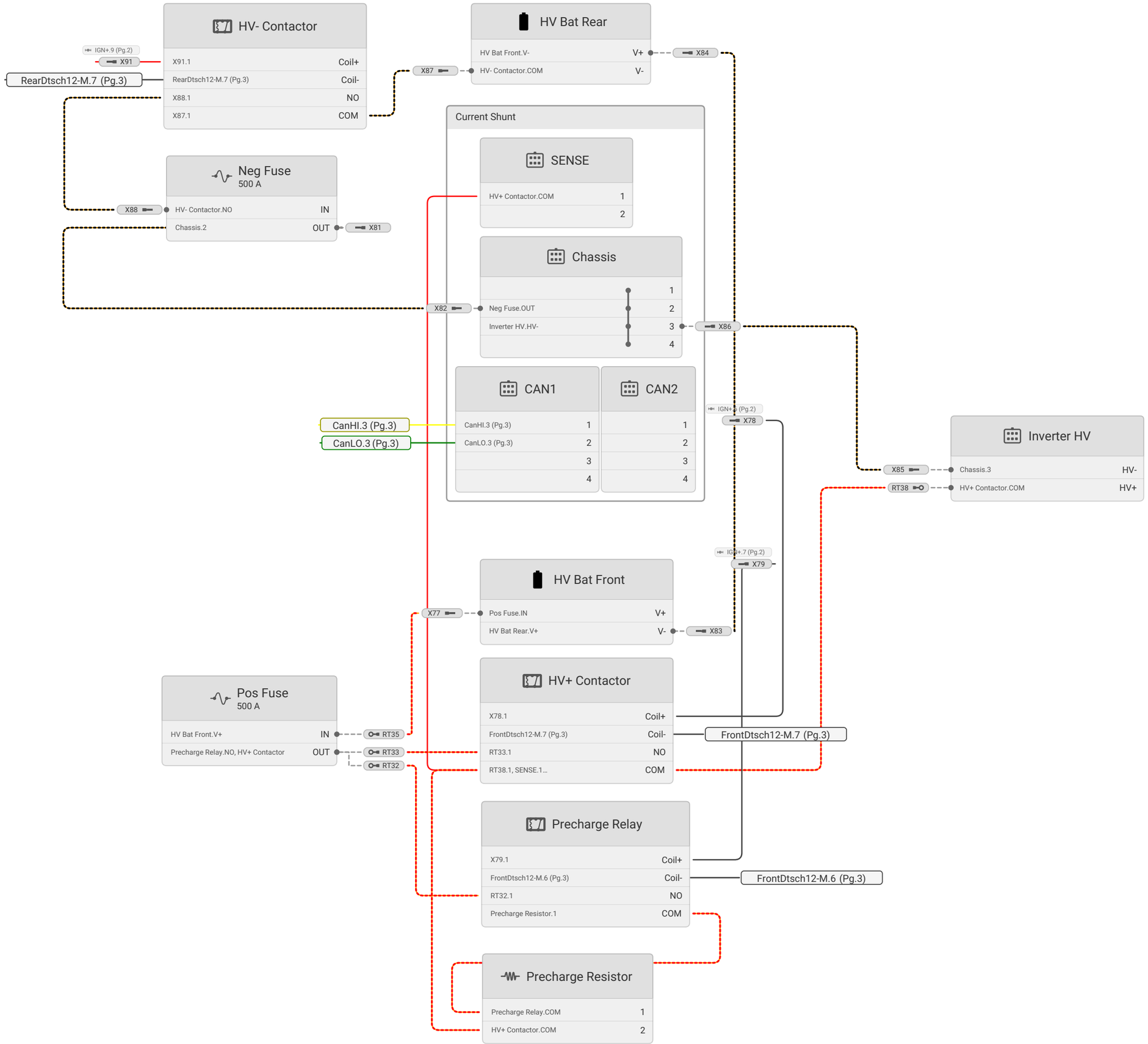


Battery box LV — Layout


DCDC-R
12-5 DCDC
RearDtsch12-M.1 Gnd 1
RearDtsch12-M.12 +12 2
BMS-R.2 +5 3
BMS-R.1 Gnd 4
BMS-R
BMS Slaves
DCDC-R.4 Gnd 1
DCDC-R.3 +5 2
RearDtsch12-M.4 CanHI 3
RearDtsch12-M.5 CanLo 4
RearDtsch12-M
Rear BatBox Signal connector
DCDC-R.1 Gnd 1
CANHi-in 2
CANLO-in 3
BMS-R.3 CANHI-Return 4
BMS-R.4 CANLO-Return 5
PreCharge 6
HV- Contactor.Coil- (Pg.4) Contactor 7
Welded+ 8
Welded- 9
ISA1+ 10
ISA1- 11
DCDC-R.2 BAT+ 12
RearDtsch12-F
Rear BatBox Signal conn...
FrontDtsch12 Gnd 1
FrontDtsch12 CANHi-in 2
FrontDtsch12 CANLO-in 3
FrontDtsch12 CANHI-R... 4
FrontDtsch12 CANLO-R... 5
FrontDtsch12 PreChar... 6
FrontDtsch12 Contact... 7
FrontDtsch12 Welded+ 8
FrontDtsch12 Welded- 9
FrontDtsch12 ISA1+ 10
FrontDtsch12 ISA1- 11
FrontDtsch12 BAT+ 12
WG1
FrontDtsch12-F
Front BatBox Signal con...
RearDtsch12 Gnd 1
RearDtsch12 CANHi-in 2
RearDtsch12 CANLO-in 3
RearDtsch12 CANHI-R... 4
RearDtsch12 CANLO-R... 5
RearDtsch12 PreChar... 6
RearDtsch12 Contact... 7
RearDtsch12 Welded+ 8
RearDtsch12 Welded- 9
RearDtsch12 ISA1+ 10
RearDtsch12 ISA1- 11
RearDtsch12 BAT+ 12
FrontDtsch12-M
Rear BatBox Signal connector
DCDC-F.1 Gnd 1
CANHi-in 2
CANLO-in 3
CanHI.1 CANHI-Return 4
CanLO.1 CANLO-Return 5
Precharge Relay.Coil- (Pg.4) PreCharge 6
HV+ Contactor.Coil- (Pg.4) Contactor 7
Welded+ 8
Welded- 9
ISA1+ 10
ISA1- 11
DCDC-F.2 BAT+ 12
DCDC-F
12-5 DCDC
FrontDtsch12-M.1 Gnd 1
FrontDtsch12-M.12 +12 2
BMS-F.2 +5 3
BMS-F.1 Gnd 4
BMS-F
BMS Slaves
DCDC-F.4 Gnd 1
DCDC-F.3 +5 2
CanHI.2 CanHI 3
CanLO.2 CanLo 4
CanHI
CanHI
FrontDtsch12-M.4 1
BMS-F.3 2
CAN1.1 (Pg.4) 3
4
5
6
CanLO
CanLO
FrontDtsch12-M.5 1
BMS-F.4 2
CAN1.2 (Pg.4) 3
4
5
6
HV- Contactor.Coil- (Pg.4)
HV+ Contactor.Coil- (Pg.4)
Precharge Relay.Coil- (Pg.4)
CAN1.1 (Pg.4)
CAN1.2 (Pg.4)



HV Batteries — Layout


HV- Contactor
IGN+.9 (Pg.2)
X91
X91.1
Coil+
RearDtsch12-M.7 (Pg.3)
RearDtsch12-M.7 (Pg.3)
Coil-
X88.1
NO
X87.1
COM
HV Bat Rear
HV Bat Front.V-
V+
X84
X87
HV- Contactor.COM
V-
Current Shunt
SENSE
HV+ Contactor.COM
1
2
Neg Fuse
500 A
X88
HV- Contactor.NO
IN
Chassis.2
OUT
X81
Chassis
1
X82
Neg Fuse.OUT
2
Inverter HV.HV-
3
X86
4
CAN1
CAN2
IGN+.9 (Pg.2)
X78
CanHI.3 (Pg.3)
CanLO.3 (Pg.3)
CanHI.3 (Pg.3)
1
1
CanLO.3 (Pg.3)
2
2
3
3
4
4
Inverter HV
X85
Chassis.3
HV-
RT38
HV+ Contactor.COM
HV+
IGN+.7 (Pg.2)
X79
HV Bat Front
X77
Pos Fuse.IN
V+
HV Bat Rear.V+
V-
X83
HV+ Contactor
Pos Fuse
500 A
X78.1
Coil+
HV Bat Front.V+
IN
RT35
FrontDtsch12-M.7 (Pg.3)
FrontDtsch12-M.7 (Pg.3)
Coil-
Precharge Relay.NO, HV+ Contactor
OUT
RT33
RT33.1
NO
RT32
RT38.1, SENSE.1...
COM
Precharge Relay
X79.1
Coil+
FrontDtsch12-M.6 (Pg.3)
FrontDtsch12-M.6 (Pg.3)
Coil-
RT32.1
NO
Precharge Resistor.1
COM
Precharge Resistor
Precharge Relay.COM
1
HV+ Contactor.COM
2